

COMMUTATIVE ALGEBRA

Manish Kumar, notes by Ramdas Singh

Fifth Semester

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Chapter 1

Commutative Algebra

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Topics to be covered in this course include tensor products, projective and flat modules, primary decomposition of ideals, Hilbert Nullstellensatz, and completions. The primary references for this course are

1. Atiyah and Macdonald, *Introduction to Commutative Algebra*,
2. Eisenbud, *Commutative Algebra with a View Toward Algebraic Geometry*, and
3. Miles Reid, *Undergraduate Commutative Algebra*.

The mid-term will account for 40% of the final grade, and the final exam will account for 60% of the final grade.

1.1 Review

A *ring* here will always refer to a commutative ring with unity. An *ideal* $I \subseteq R$ of a ring R is a subset of R such that I is an additive subgroup of R and for all $r \in R$ and $x \in I$, we have $rx \in I$. A *prime ideal* $P \subseteq R$ is an ideal such that if $xy \in P$ for some $x, y \in R$, then either $x \in P$ or $y \in P$. A *maximal ideal* $\mathfrak{m} \subsetneq R$ is an ideal such that if $\mathfrak{m} \subseteq I \subseteq R$ for some ideal I , then either $I = \mathfrak{m}$ or $I = R$.

Proposition 1.1. *Every maximal ideal is a prime ideal.*

Proof. Let $\mathfrak{m}$ be a maximal ideal of R . Suppose $x, y \notin \mathfrak{m}$ for some $x, y \in R$. Then the ideals $(\mathfrak{m}, x)$ and $(\mathfrak{m}, y)$ are strictly larger than $\mathfrak{m}$, so by maximality, we have $(\mathfrak{m}, x) = R$ and $(\mathfrak{m}, y) = R$. Thus, there exist $a, b \in R$ and $m, n \in \mathfrak{m}$ such that $ax + m = 1$ and $by + n = 1$. Multiplying these two equations gives $(ax + m)(by + n) = abxy + anx + bmy + mn = 1$. Since $mn, anx, bmy \in \mathfrak{m}$, we have $abxy \in \mathfrak{m}$, showing $xy \notin \mathfrak{m}$. Therefore, $\mathfrak{m}$ is a prime ideal. ■

Proposition 1.2. *Let R be a ring and $I \subseteq R$ be an ideal. Then I is a prime ideal if and only if the quotient ring R/I is an integral domain.*

Proof. Suppose I is a prime ideal. Let $a + I, b + I \in R/I$ be such that $(a + I)(b + I) = 0 + I$. Then $ab + I = 0 + I$, so $ab \in I$. Since I is a prime ideal, either $a \in I$ or $b \in I$. Thus, either $a + I = 0 + I$ or $b + I = 0 + I$, showing that R/I is an integral domain. A similar argument holds for the converse. ■

Recall that a ring R is called a *field* if $R \setminus \{0\}$ is a group under multiplication.

Proposition 1.3. *Let R be a ring and $I \subseteq R$ be an ideal. Then I is a maximal ideal if and only if the quotient ring R/I is a field.*

Proof. Suppose I is a maximal ideal. Let $a + I \in R/I$ be such that $a + I \neq 0 + I$. Then $a \notin I$, so the ideal (I, a) is strictly larger than I . By maximality, we have $(I, a) = R$. Thus, there exist $r \in R$ and $m \in I$ such that $ra + m = 1$. Then $(a + I)(r + I) = ra + I = 1 + I$, showing that $a + I$ is invertible. Therefore, R/I is a field. A similar argument holds for the converse. ■

Proposition 1.4. *Let R be a ring and $I \subseteq R$ be an ideal. Then the ideals in R containing I are in one-to-one correspondence with the ideals in the quotient ring R/I .*

A *zero-divisor* in a ring R is an element $x \in R$ such that there exists a non-zero element $y \in R$ with $xy = 0$. A *nilpotent* in a ring R is an element $x \in R$ such that there exists a positive integer n with $x^n = 0$. A ring with no non-zero zero-divisors is called an *integral domain*. A ring with no non-zero nilpotents is called a *reduced ring*.

The *nilradical* of a ring R is the subset (ideal) of all nilpotent elements in R , denoted by $\text{nil}(R)$ or $\sqrt{(0)}$. The *Jacobson radical* of a ring R is the intersection of all maximal ideals in R , denoted by $\text{Jac}(R)$. The *radical* of an ideal $I \subseteq R$ is the set of all elements $x \in R$ such that $x^n \in I$ for some positive integer n , denoted by $\text{rad}(I)$ or $\sqrt{I}$.

Proposition 1.5. *Let R be a ring and $I \subseteq R$ be an ideal, and $q : R \rightarrow R/I$ be the quotient map. Then $\sqrt{I} = q^{-1}(\text{nil}(R/I))$.*

Proof. Let $x \in \sqrt{I}$. Then $x^n \in I$ for some positive integer n , so $q(x^n) = 0 + I$. Thus, $q(x)^n = 0 + I$, showing that $q(x)$ is nilpotent in R/I . Therefore, $x \in q^{-1}(\text{nil}(R/I))$. Conversely, let $x \in q^{-1}(\text{nil}(R/I))$. Then $q(x)$ is nilpotent in R/I , so there exists a positive integer n such that $q(x)^n = 0 + I$. Thus, $q(x^n) = 0 + I$, so $x^n \in I$, showing that $x \in \sqrt{I}$. Therefore, $\sqrt{I} = q^{-1}(\text{nil}(R/I))$. ■

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